UNIVERSITY OF GREENWICH
FACULTY OF ENGINEERING AND SCIENCE
SCHOOL OF COMPUTING AND MATHEMATICAL SCIENCES

COMP1682 Final Year Project Proposal

Heart Disease Prediction using Medical Data

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BSc (Hons) COMP-1682-Final Year Project

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DECLARATION OF AI USE

I have used AI while undertaking my assignment in the following ways:

- To develop research questions on the topic – NO
- To create an outline of the topic – NO
- To explain concepts – YES
- To support my use of language – YES
- To summarise the following articles/resources: – YES
    1. Jyoti Soni - Predictive Data Mining for Medical Diagnosis
    2. Harshit Jindal - Heart disease prediction using machine learning algorithms
    3. Senthilkumar Mohan - Effective Heart Disease Prediction Using Hybrid Machine Learning Techniques
    4. M. Abinaya - Heart Disease Prediction using Machine Learning and Data Analytics Approach
    5. V.V. Ramalingam - Heart disease prediction using machine learning techniques : a survey
    6. Lundberg, S.M. & Lee, S.-I. - A Unified Approach to Interpreting Model Predictions

In other ways, as described below: – NO

SYNOPSIS

Cardiovascular Diseases (CVD) is the leading cause of mortality worldwide and claimed almost 17.9 million deaths annually (World Health Organization, 2024). Progress continues to be slow even with endless attempts to improve prediction and prevention accuracy due to the heavy reliance on linear diagnostic techniques which do not capture the complex interdependence of cholesterol, age, and blood pressure (Acharjya & Anand, 2021).

This project aims to improve the precision of heart-disease detection using machine learning (ML) and deep learning (DL). A combination predictive model of Random Forest, XGBoost, and a feed-forward Artificial Neural Network (ANN) will be used to uncover risk patterns in patient records (Breiman 2001; Chen & Guestrin 2016; LeCun/Bengio/Hinton 2015).

SHAP and LIME will be employed to track model authorship, enabling model transparency by accounting for each clinical variable in prediction outcomes (Lundberg & Lee 2017; Ribeiro/Singh/Guestrin 2016).

Accuracy, ROC-AUC, recall, and precision will be used to gauge model performance on the Heart Disease Prediction Dataset (Damarla 2024).

This research seeks to provide an ethical and explainable AI approach that enhances clinical diagnostic reliability by offering clear and reproducible predictive models.

Keywords: Cardiac illness prediction, Machine Learning, Deep Learning, Explainable AI, Interpretability

1. Aim and Objectives

1.1 Aim

This project focuses on creating a hybrid predictive system designed for the early detection of cardiovascular disease (CVD) using ensemble learning and deep learning techniques. In this study, the combination of Random Forest, XGBoost, and a feed-forward Artificial Neural Network (ANN) will be utilized to attain linear and nonlinear risk patterns in the patient data. Moreover, this research will use SHAP and LIME to interpret model predictions and provide transparency on the data to explain clinical factors associated with the outcomes of heart disease. The project aims to develop ethical and explainable artificial intelligence that will improve accuracy, trust, and reproducibility in medical diagnostics.

1.2 Objectives

This project is aimed at reviewing and finding the gaps in the interpretability and reproducibility of the current works in Cardiovascular Disease (CVD) prediction using machine learning (ML) and deep learning (DL) and explainable AI (XAI). It intends to preprocess and prepare the heart disease prediction dataset (Damarla, 2024) through cleaning, normalisation, and class balanced to aid model generalisation using the SMOTE technique. A hybrid model combining Random Forest and XGBoost with a feed-forward Artificial Neural Network (ANN) will be built to ensure accurate and stable disease prediction. The study will rigorously use SHAP and LIME to explain global and local model features to ensure proper understanding of model-aided decisions. The models will be evaluated and benchmarked against the baseline Logistic Regression model to validate performance improvements using multiple metrics like accuracy, recall, precision, and ROC-AUC on explainable models. Finally, the project will address compliance with the GDPR (European Union, 2018) and the University of Greenwich Research Ethics Policy (2025) with emphasis on reproducibility and ethical compliance.

2 BACKGROUND RESEARCH AND PROJECT RATIONALE

2.1 Early Research on Machine Learning for CVD Prediction

Early studies focused on the application of classical machine learning models with promising outcomes for the prediction of cardiovascular diseases. Soni et al. (2011) and Pattekari & Parveen (2012) used Decision Trees and Naïve Bayes models and proved their accuracy was higher than that of manual diagnostic assessments thereby indicating the potential of automated, data driven prediction. Unfortunately, the studies used small and balanced datasets, which are inconsistently scalable to real world clinical settings. Algorithm performance on the data used in Kumar & Singh (2013) pointed towards poor preprocessing and attribute selection and the unexplored possibilities of deeper learning or ensemble integration of algorithms.

2.2 Advances in Ensemble and Hybrid Models

The growing body of research increasingly uses ensemble techniques in predicting heart disease. Ramalingam et al. (2018) conducted a survey of the most advanced ML techniques and noted recurring issues such as class imbalance, overfitting, and lack of reproducibility. To improve predictive reliability, the studies conducted by Mohan et al. (2019) and Ghorpade and Sirsikar (2021) implemented ensemble models, particularly Random Forest, and noted the issues of interpretability and clinical transparency that remained unsolved. Abinaya and Karthikeyan (2020) identified the potential for improved predictive performance stemming from preprocessing and feature engineering but their work was limited by a narrow dataset and therefore lack of generalisability. Overall, these studies report that important advances in algorithms and greater explainability and ethically relevant use remain to be developed.

2.3 Explainability and and Ethical AI in Healthcare

In healthcare applications, the interpretability of AI models is being considered a major issue. Lundberg & Lee (2017) published the method SHAP, which aims to measure the global contributions of the different features in predictive models. The more locally focused method LIME, which aims to explain individual predictions, was introduced by Ribeiro, Singh & Guestrin (2016). While these approaches built some level of transparency into the models, studies of data for the years of 2019 and 2022, such as Floridi et al. and Obermeyer et al., demonstrated the absence of fairness, accountability, and reproducibility, in which case the lack of these elements was a major issue. Wang & Patel (2023) underscored that many clinical AI models state their accuracy alongside a lack of explainability and ethical scrutiny, signifying the importance of reproducible and interpretable solutions.

2.4 Project Rationale

This project aims to fill these gaps. A hybrid predictive framework that combines Random Forest, XGBoost, and a feed-forward Artificial Neural Network (ANN) is being developed for the purpose of learning both linear and nonlinear relationships for the cardiovascular dataset. To achieve transparency and foster trust, SHAP and LIME will be used for model explainability. This directly addresses the opaque and non-reproducible systems described in the earlier studies. The goal is to achieve a technical performance that is ethically and clinically usable, promoting the goals of explainable responsible AI in healthcare.

3 METHODOLOGY AND TECHNICAL APPROACH

This project takes a systematic, data-driven approach to designing and assessing a hybrid deep learning and ensemble techniques system for the prediction of cardiovascular diseases (CVD). The project is also guided primarily by the CRISP-DM framework, moving forward with understanding the data, followed by modelling and evaluating the systems in a repetitive fashion.

3.1 Data Collection And Processing

The study will utilize the Heart Disease Prediction Dataset (Damarla, 2024) available on Kaggle. Data preprocessing will involve cleaning, normalization, feature engineering, and class imbalance correction using SMOTE for better generalization. Exploratory analysis will

focus on determining the most important predictive variables, including age, cholesterol, and resting blood pressure.

3.2 Model Development

The Random Forest, XGBoost, and feed-forward Artificial Neural Network (ANN) each will be constructed, and then compared to one another. Hyperparameter training and tuning will be addressed using GridSearchCV. Several hidden layers each of which will be equipped with ReLU activations, coupled with dropout regularisation, and optimised with Adam will be used for the ANN.

3.3 Integration and Web Interface (Application Development)

As part of the system prototype to enhance practical utility, a web-based application will be developed on a full stack. This will allow users to add a patient's particulars and get the prediction results instantaneously from the deployed hybrid model. This part of the project shows how AI can be integrated into smart and easy-to-use health care technology, and adds uniqueness to the project.

3.4 Model Evaluation and Explainability

The model will be evaluated on its accuracy, precision, recall, F1-score, and ROC–AUC metrics. Explanations of global and local feature contributions will be accomplished using SHAP and LIME, respectively, focusing on clinical explainability. A baseline Logistic Regression model will be used for comparison to assess the model's relative improvement.

3.5 Tools and Technologies

For implementation, we will make use of Python, Tensorflow, scikit-learn, Pandas and use Flask for web-deployment. Like all other projects, this one will also observe ethical standards and reproducibility as outlined in the GDPR (European Union, 2018) and the University of Greenwich's Research Ethic Policy (2025).

4 EVALUATION PLAN

The evaluation will determine the accuracy, reliability, and interpretability of the proposed hybrid model for cardiovascular disease (CVD) prediction. Both quantitative and qualitative analyses will be used to assess the performance of the machine learning (ML) and deep

learning (DL) models, ensuring that the study provides a balanced evaluation of predictive quality, transparency, and reproducibility.

4.1 Quantitative Assessment

The hybrid model will be evaluated on several metrics regarding it's predictive performance and robustess. This will be done quantitatively.

4.1.1 Accuracy: This indicator kollects the correctness of transparent outcomes prediction and accordingly assesses the trust of the model in question.

4.1.2 Precision and Recall: Precision deals with the ratio of the predicted cases of positive condition which were classified as such correctly, and recall assesses how many of the actual positive cases were identified and classified correctly.

4.1.3 F1-score: The degree of the weighted mean of precision and recall which gauges the balance between sensitivity and specificity.

4.1.4 ROC-AUC: The area under the Receiver Operating Characteristic curve will be the figure that represents the model's ability to distinguish the patients to the categories of healthy and sick.

4.1.5 Training Time and Complexity: Alongside the models, the loss of each will be recorded to measure the tradeoff of efficiency and precision (Wang & Patel, 2023).

Cross validation will be executed while using the stratified k-fold validation method to ensure the results will be as consistent and unbiased as possible. To assess performance gains and overfitting hybrid models will be evaluated against a baseline Logistic Regression model.

4.2 Qualitative Assessment

Qualitative evaluation will focus on interpretability and ethical performance. Both global and local feature contributions will be explained using SHAP (Lundberg & Lee 2017) and LIME (Ribeiro et al 2016) methodologies elucidating how clinical parameters such as age, cholesterol, and blood pressure shape clinical predictions.

Model predictions will be explicated biomedically, checked for visual alignment through plots and feature-importance comparisons. The model outputs will be checked (if suggested) against a body of a literature and insights from subject matter specialists to ensure that model

predictions will be clinically meaningful, reproducible, and ethically compliant with GDPR (European Union 2018).

5 LSEPI CONSIDERATIONS

The project remains compliant with all legal, social, ethical, and professional aspects of managing medical data and developing ai systems.

5.1 Legal

The project is compliant with UK GDPR and Data Protection Act (2018). The only data set that will be used is the Heart Disease Prediction Dataset (Damarla, 2024) which is public and anonymised. No personally identifiable information will be collected, and all files will be securely stored and deleted upon project completion in accordance with University of Greenwich Research Ethics Policy (2025).

5.2 Social

The project fosters the use of responsible and trustworthy ai in healthcare by encouraging transparency in medical predictions. It does not permit any clinical misuse of results and aims to strengthen the trust in ai systems' diagnoses by providing clear and interpretable results.

5.3 Ethical

Approval from the Faculty of Engineering and Science Ethics Committee will be sought prior to execution of the project. The research does not breach the principles of ethical participation and data collection. The absence of human subjects or personal sensitive data ensures that the principles of ethical use of the IEEE (2020) and the ACM (2018) professional code is respected.

5.4 Professional

The entire process of conducting the experiments alongside the code and findings will be done and reported honestly, ensuring total replicability and ethical professional oversight in the entire project.

6 RISK ASSESSMENT

This project is perceived as low-risk but has potential issues regarding data security, model efficacy, and technical dependability. The table below describes the principal risks as well as the solutions employed to keep the effect to a minimum, all while upholding ethical and professional paradigms.

Potential Risk	Likelihood	Impact	Mitigation Strategy
Data breach or unauthorised access	Low	Medium	All datasets are publicly available and anonymised. Data and code will be stored on the University's encrypted OneDrive with restricted access to the researcher and supervisor.
Data loss or corruption	Medium	Medium	Regular backups will be maintained on secure cloud storage. All scripts and results will be version-controlled using GitHub.
Technical failure (model or system error)	Medium	Medium	Testing will be conducted before training and deployment to identify errors. Computational tasks will be verified on local and cloud environments.
Ethical or compliance oversight	Low	High	The project will follow UK GDPR and University research ethics policies. Ethical self-assessment will be completed before implementation.
Model underperformance	Medium	Medium	Model tuning through cross-validation and parameter optimisation will reduce bias and overfitting. Results will be reviewed regularly with the supervisor.

7 PROJECT PLAN AND TIMELINE

The project is organised into five sequential stages. Each stage lists core activities, estimated duration, the standard cohort timeframe, and the main deliverables.

Stage	Core activities	Duration	Timeframe	Deliverables / Outputs
Proposal & Ethics	Define aim and objectives, conduct preliminary literature review, and prepare ethics and risk documentation.	4 weeks	Oct – Nov 2025	Final project proposal + ethics submission pack
Literature, Data & Design	Complete literature review, finalise methodology, obtain and preprocess dataset, and prepare data summary.	8 weeks	Nov 2025 – Jan 2026	Contextual report + cleaned dataset summary
Model Development & Prototype	Implement Random Forest, XGBoost, and ANN; perform hyperparameter tuning; and build a simple web prototype.	6 weeks	Jan – Mar 2026	Working model prototypes, code repository, simple web prototype
Evaluation & Analysis	Conduct cross-validation, measure accuracy and complexity, apply SHAP/LIME, and compare model results.	6 weeks	Mar – Apr 2026	Performance report, SHAP/LIME figures, reproducible notebooks
Documentation & Demonstration	Finalise report, prepare figures, and complete presentation and demonstration materials.	1 weeks	23–24 Apr 2026	Final project submission + demonstration materials

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